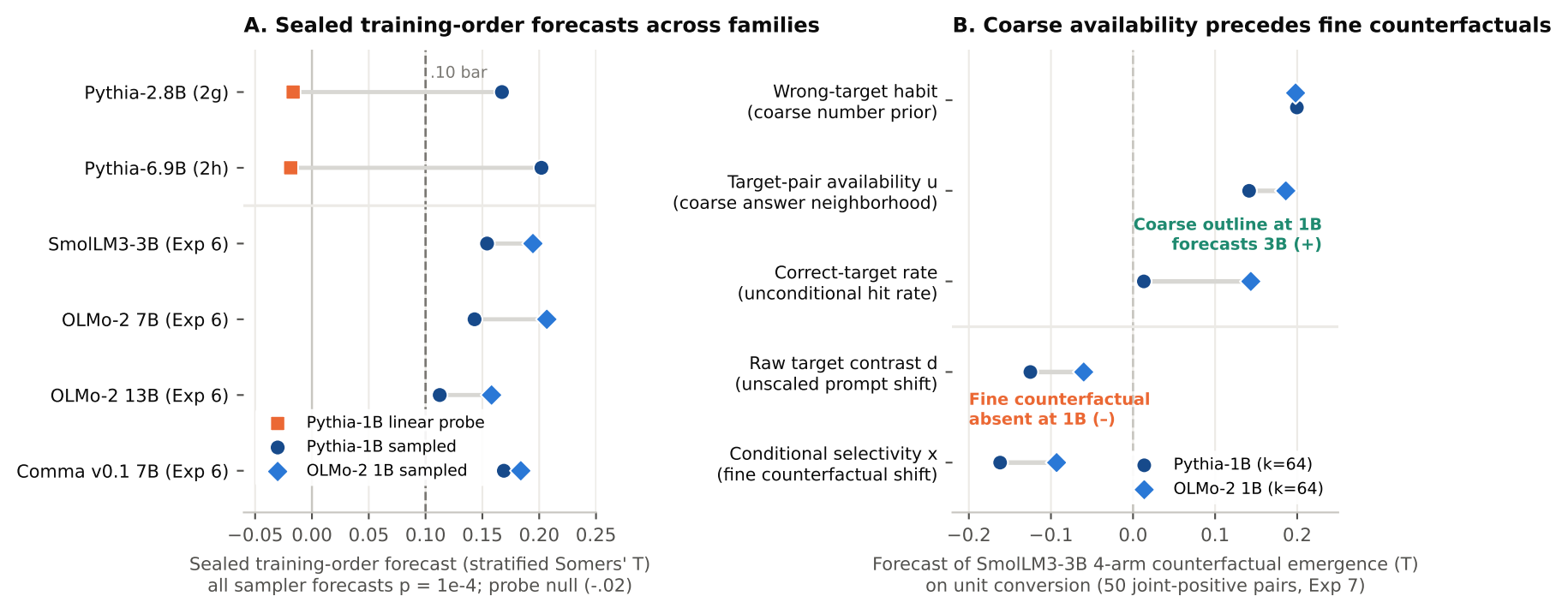
A. Sealed training-order forecasts across families
.10 bar
Pythia-2.8B (2g)
Pythia-6.9B (2h)
SmolLM3-3B (Exp 6)
OLMo-2 7B (Exp 6)
OLMo-2 13B (Exp 6)
Comma v0.1 7B (Exp 6)
Pythia-1B linear probe
Pythia-1B sampled
OLMo-2 1B sampled
−0.05 0.00 0.05 0.10 0.15 0.20 0.25
Sealed training-order forecast (stratified Somers' T)
all sampler forecasts p = 1e-4; probe null (-.02)
B. Coarse availability precedes fine counterfactuals
Wrong-target habit
(coarse number prior)
Target-pair availability u
(coarse answer neighborhood)
Correct-target rate
(unconditional hit rate)
Raw target contrast d
(unscaled prompt shift)
Conditional selectivity x
(fine counterfactual shift)
Coarse outline at 1B
forecasts 3B (+)
Fine counterfactual
absent at 1B (–)
Pythia-1B (k=64)
OLMo-2 1B (k=64)
−0.2 −0.1 0.0 0.1 0.2
Forecast of SmolLM3-3B 4-arm counterfactual emergence (T)
on unit conversion (50 joint-positive pairs, Exp 7)